

Semester Test 3 / Aegrotat Test

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Electricity and Electronics EBN111

28 May 2025

Assessment Information

Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR answer sheet completion)	Open/Closed book:	Closed
Total number of pages (this page included)		16	

Lecturers: Prof X. Ye, Mr S. Twala, Mr A. Rohde**Assistant Lecturers:** L. Hurter, J. Nepembe, I. Schilz, C. van Zyl**Important**

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of the OCR answer sheet 00/01.
4. Read the instructions on the OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. The final answers must be written as real numbers and not as fractions.
7. Use at least 4 significant figures to write numbers, including complex numbers.
8. A complex number may be written in either rectangular or polar format.
9. Exact numbers in which decimal expansions are zero can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

INSTRUCTIONS

STEP 1 (-5 min to 0 min): Complete the front page of OCR answer sheet 01, and also fill in your student number on additional OCR answer sheets.

STEP 2 (0 min to 90 min): Do your calculations on the question paper.

STEP 3 (90 min to 100 min): Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

- The Assessment ID is: **2025EBN111S03**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should have only one character, and the characters should not touch or cross the cell.
- **Do not use commas!** Please use full stops as decimal points.

Examples for complex numbers on OCR forms

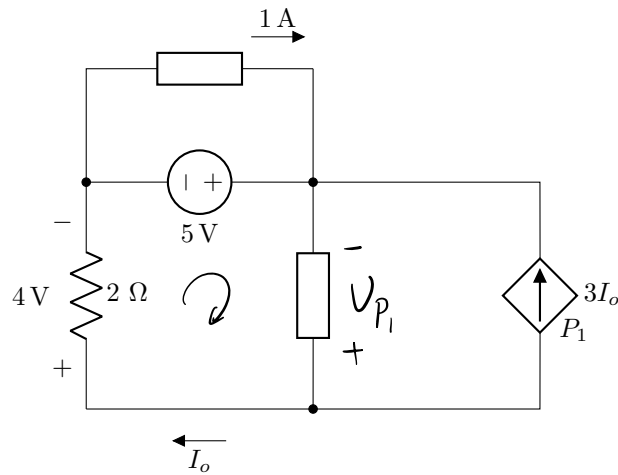
5	∠	—	9	0	°				V	
6	.	9	∠	1	5	.	9	°	μ	A
1	2	—	j	8						
j	1	5							Ω	
—	1	.	5	+	j	8	.	2	m	V

Rectangular:The j is at the start of the imaginary number.The j is in the CENTRE of the cell.The j does NOT cross or touch cell boundaries!**Polar:**

The degree sign ° is directly after the angle value.

Section 1

Question 1 [2]

Find the power associated with the dependent current source P_1 .01 P_1 [2] -6 W

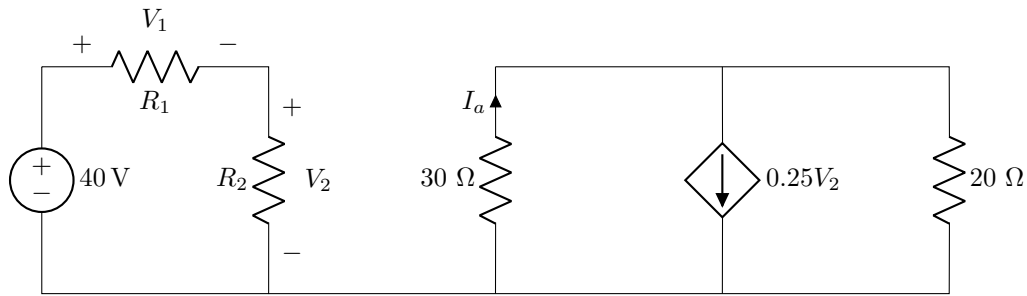
$$4 - 5 - V_{P_1} = 0 \Rightarrow V_{P_1} = -1\text{ V.}$$

$$I_o = \frac{4}{2} = 2\text{ A}, \quad 3I_o = 6\text{ A.}$$

$$\textcircled{01} \quad P_1 = V_{P_1} \cdot 3I_o = -6\text{ W.}$$

Questions 2 to 3 [4]

Consider the circuit below and assume that $V_1/V_2 = 3$ and $R_2 = 4 \Omega$.



02 Calculate R_1 . [2] 12Ω

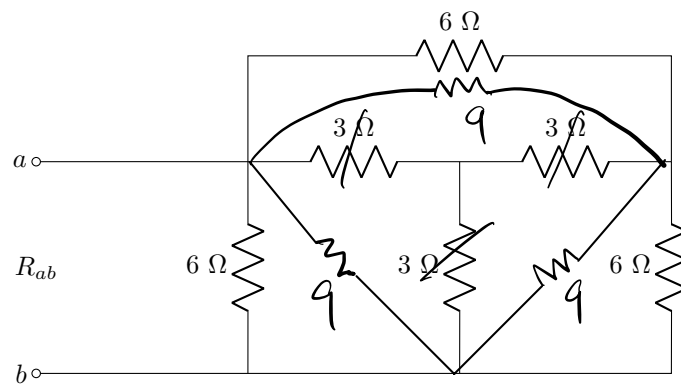
03 Calculate I_a . [2] 1 A

$$\textcircled{02} \quad \frac{V_1}{V_2} = \frac{R_1}{R_2} = 3 \Rightarrow \underline{R_1 = 12 \Omega.}$$

$$\textcircled{03} \quad V_2 = 10 \text{ V},$$

$$I_a = 0.25V_2 \cdot \frac{20}{20+30} = 2.5 \cdot \frac{2}{5} = 1 \text{ A}$$

Question 4 [2]



04

Determine the equivalent resistance R_{ab} between nodes a and b . [2] 2.4Ω

$$R_{ab} = (6 \parallel 9) \parallel (6 \parallel 9 + 6 \parallel 9)$$

$$= \frac{6 \times 9}{6 + 9} \parallel \frac{6 \times 9 \times 2}{6 + 9}$$

$$= 3.6 \parallel 7.2 = \frac{3.6 \times 7.2}{3.6 + 7.2} = 2.4 \Omega.$$

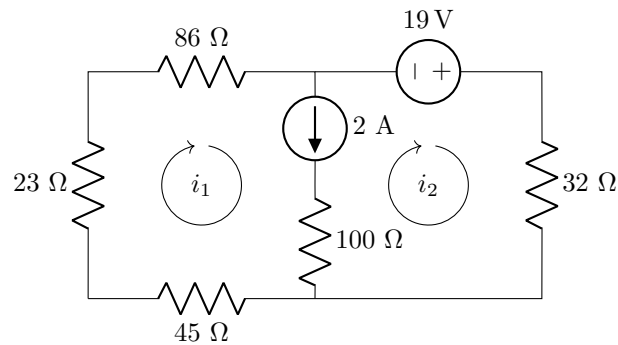
Questions 5 to 8 [4]

Use mesh analysis to set up the following two equations of the form:

$$ai_1 + bi_2 = 19$$

$$ci_1 - i_2 = d$$

Calculate the unknown parameters of a , b , c and d .



$$i_1 - i_2 = 2$$

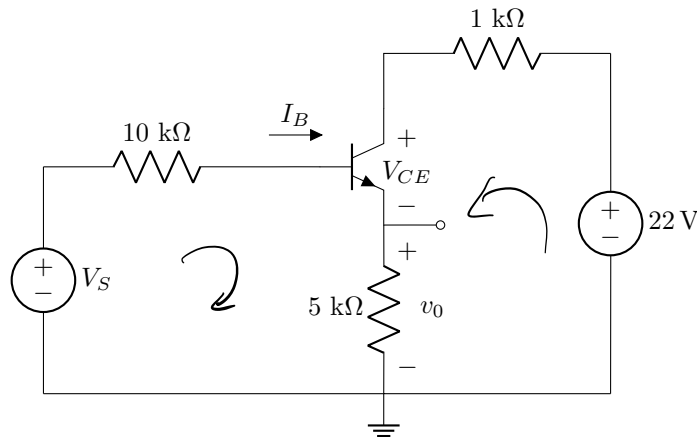
05	a [1]	$= 154$
06	b [1]	$= 32$
07	c [1]	$= 1$
08	d [1]	$= 2$

Super mesh:

$$(23 + 86 + 45)i_1 - 19 + 32i_2 = 0$$

$$154i_1 + 32i_2 = 19$$

Questions 9 to 10 [4]

Assume that the transistor operates in the active region, and $\beta = 99$:09 Calculate I_B when $V_S = 5$ V. [2] 8.43 μ A10 Calculate the voltage V_{CE} if $v_0 = 5$ V. [2] 16.01 V

$$(09) -V_S + 10 \times 10^3 \cdot I_B + 5 \times 10^3 (I_B + \beta I_B) + V_{BE} = 0$$

$$(10 \times 10^3 + 5 \times 10^3 \times 100) I_B = 5 - 0.7$$

$$I_B = \frac{5 - 0.7}{(10 + 500) \times 10^3} = 8.43 \mu A$$

$$(10) V_0 = (1 + \beta) I_B \times 5 \times 10^3$$

$$I_B = \frac{5}{100 \times 5 \times 10^3} = 10 \mu A$$

$$-22 + 10^3 I_C + V_{CE} + 5 = 0$$

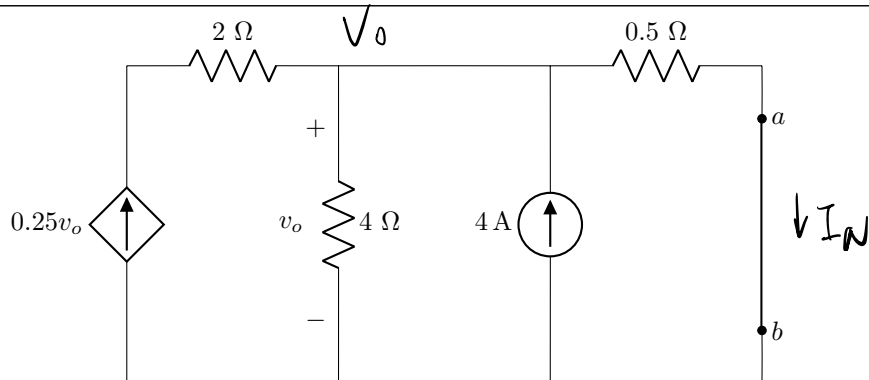
$$V_{CE} = 22 - 5 - 10^3 \cdot 99 \cdot 10 \times 10^{-6} \\ = 16.01 \text{ V.}$$

Total Section 1: [16]

Section 2

Questions 11 to 12 [4]

Find the Norton equivalent current I_N , and Norton equivalent resistance R_N , with respect to terminals $a - b$.

11 I_N [2]

4A

12 R_N [2]0.5 Ω

(11)

Nodal analysis

$$-0.25V_o + \frac{V_o}{4} - 4 + \frac{V_o}{0.5} = 0$$

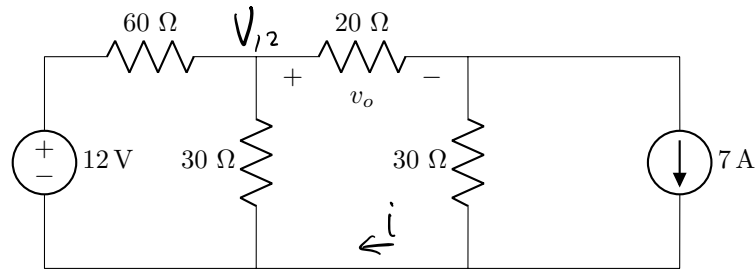
$$\Rightarrow V_o = 2V, \quad I_N = \frac{2}{0.5} = 4A$$

(12)

$$V_{Th} = V_o = 2V,$$

$$R_N = \frac{V_{Th}}{I_N} = 0.5 \Omega.$$

Questions 13 to 14 [4]

Find the individual contributions from the 12 V and 2 A sources respectively to v_o .

13	Individual contribution from 12 V source [2]	1.143V
14	Individual contribution of 2 A source [2]	60V

(13)

$$\frac{V_{12} - 12}{60} + \frac{V_{12}}{30} + \frac{V_{12}}{50} = 0 \quad \times 300$$

$$5V_{12} - 60 + 10V_{12} + 6V_{12} = 0$$

$$21V_{12} = 60 \Rightarrow V_{12} = \frac{60}{21} \text{ V}$$

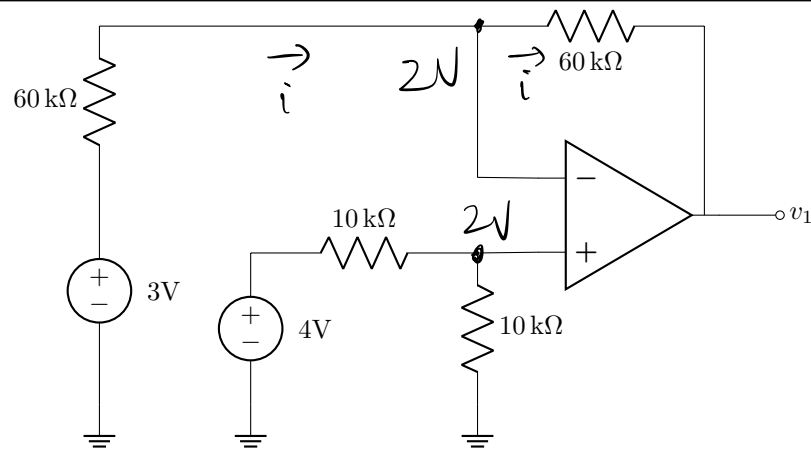
$$V_o = V_{12} \cdot \frac{2}{5} = \frac{60}{21} \cdot \frac{2}{5} = 1.143 \text{ V.}$$

(14)

$$60 \parallel 30 = 20 \Omega$$

$$i = 7 \cdot \frac{30}{40+30} = 3 \text{ A.}$$

$$V_{7A} = 3 \cdot 20 = 60 \text{ V.}$$

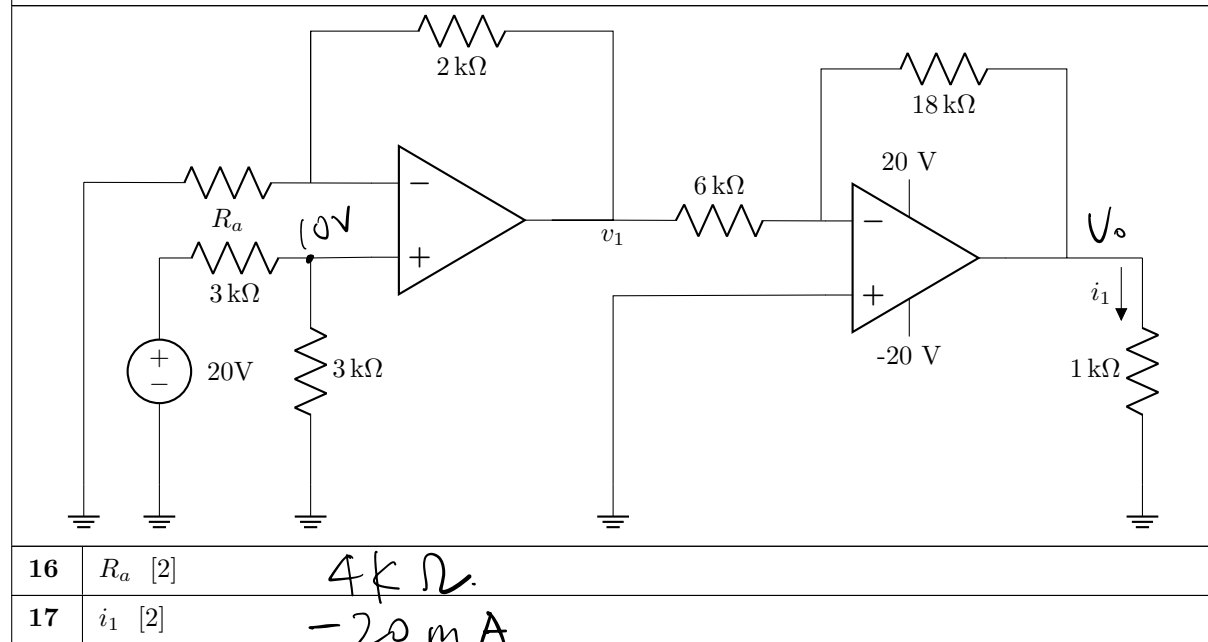
Question 15 [2]Assume that the op-amp is ideal, and find v_1 .15 v_1 [2]

1V

$$\frac{3-2}{60k} = \frac{2-v_1}{60k} \Rightarrow v_1 = 1V.$$

Questions 16 to 17 [4]

Two ideal op-amps are connected as shown. It is known that $v_1 = 15$ V, find R_a and i_1 .



$$(16) \quad v_1 = 10 \left(1 + \frac{2}{R_a} \right) = 15, \Rightarrow R_a = 4\text{ k}\Omega.$$

$$(17) \quad V_o = - \frac{18\text{ k}\Omega}{6\text{ k}\Omega} \cdot 15 = -45\text{ V}.$$

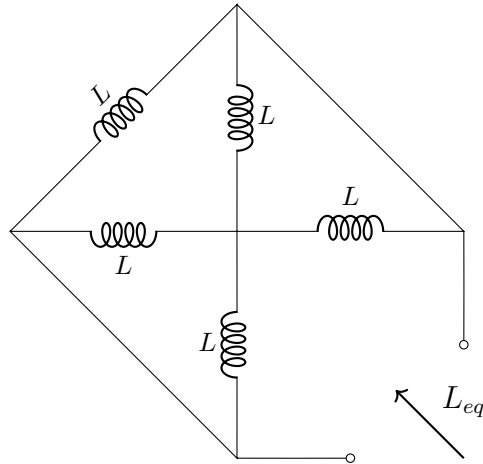
V_o can only reach -20 V due to saturation.

$$i_1 = \frac{-20}{1\text{ k}\Omega} = -20\text{ mA}.$$

Total Section 2: [14]

Section 3

Question 18 [2]

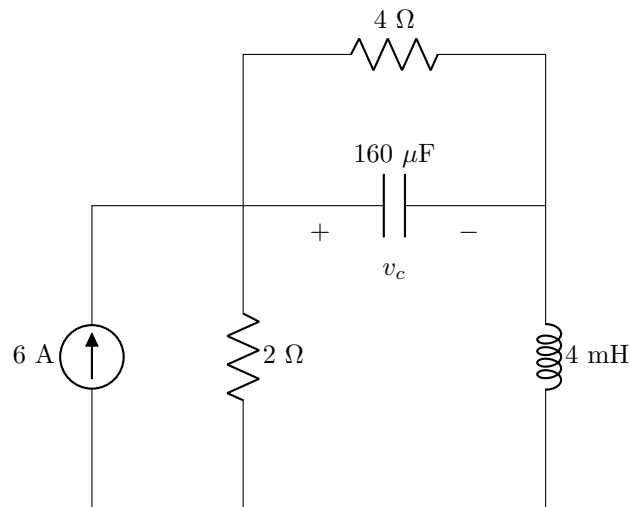
18 Find L_{eq} if $L = 10 \text{ H}$. [2]

5 H

$$L_{eq} = L \parallel (L \parallel L + L \parallel L) \\ = \frac{1}{2} L = 5 \text{ H}$$

Questions 19 to 20 [2]

The circuit below operates under steady dc conditions.



19 Find v_c . [1]

8 V

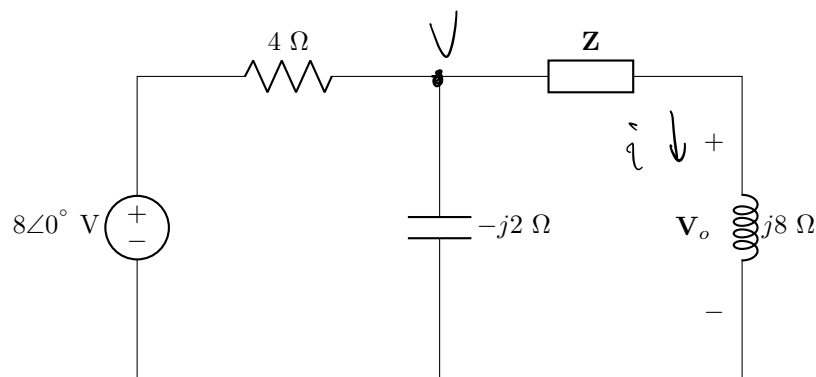
20 Find the energy stored in the capacitor. [1]

5.12 mJ

$$(19) \quad V_c = 12 \cdot \frac{4}{6} = 8\text{V.}$$

$$(20) \quad E_c = \frac{1}{2} C V_c^2 = \frac{1}{2} \times 160 \times 10^{-6} \times 64 = 5.12\text{ mJ}$$

Question 21 [2]



21

Find $\mathbf{Z}$ when $\mathbf{V}_o = 16\angle 0^\circ \text{ V}$. [2]

$$0.8 - j5.6 \, \Omega.$$

$$i = \frac{V_o}{j8} = \frac{16}{j8} = -j2 \text{ A}$$

$$\frac{V-8}{4} + \frac{V}{-j2} - j2 = 0$$

$$\underline{\times j4}$$

$$j(V-8) - 2V + 8 = 0$$

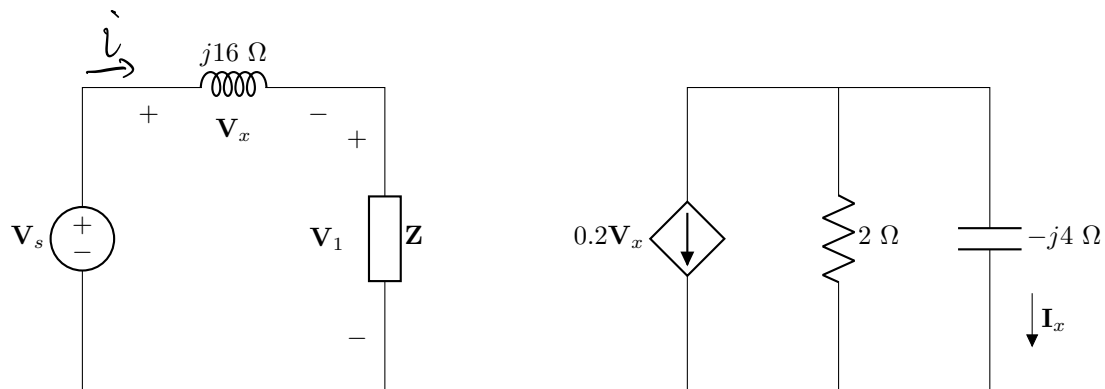
$$(-2+j)V = -8+j8$$

$$V = \frac{-8+j8}{-2+j} = 4.8 - j1.6$$

$$\frac{V}{Z+j8} = -j2 \Rightarrow Z = \frac{V}{-j2} - j8$$

$$= 0.8 - j5.6$$

Questions 22 to 23 [4]



22 Given $Z = 1 + j2 \Omega$ and $V_x = 8\angle 0^\circ$ V, find V_s . [2]

$$4 + j2 \checkmark$$

23 If $I_x = 4\angle 0^\circ$ A, find V_x . [2]

$$-20 + j40 \text{ V}$$

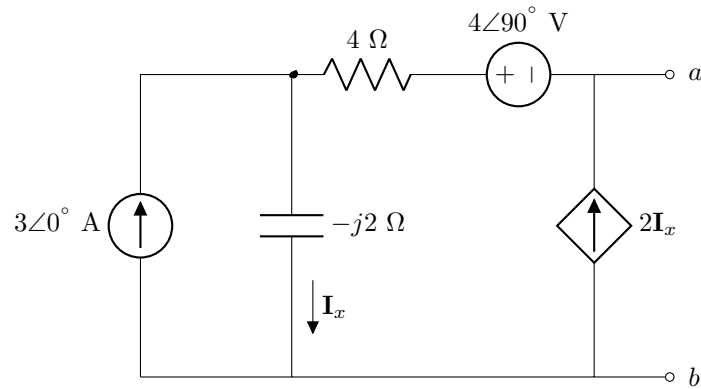
$$(22) \quad i = \frac{j16}{8} = j2, \quad V_1 = j2(1 + j2)$$

$$V_s = V_1 + V_x = j2(1 + j2) + 8 = 4 + j2 \text{ V}$$

$$(23) \quad I_x = -0.2V_x \cdot \frac{2}{2 - j4} = 4$$

$$V_x = \frac{-4(2 - j4)}{0.4} = -10(2 - j4) = -20 + j40 \text{ V}$$

Question 24 [2]

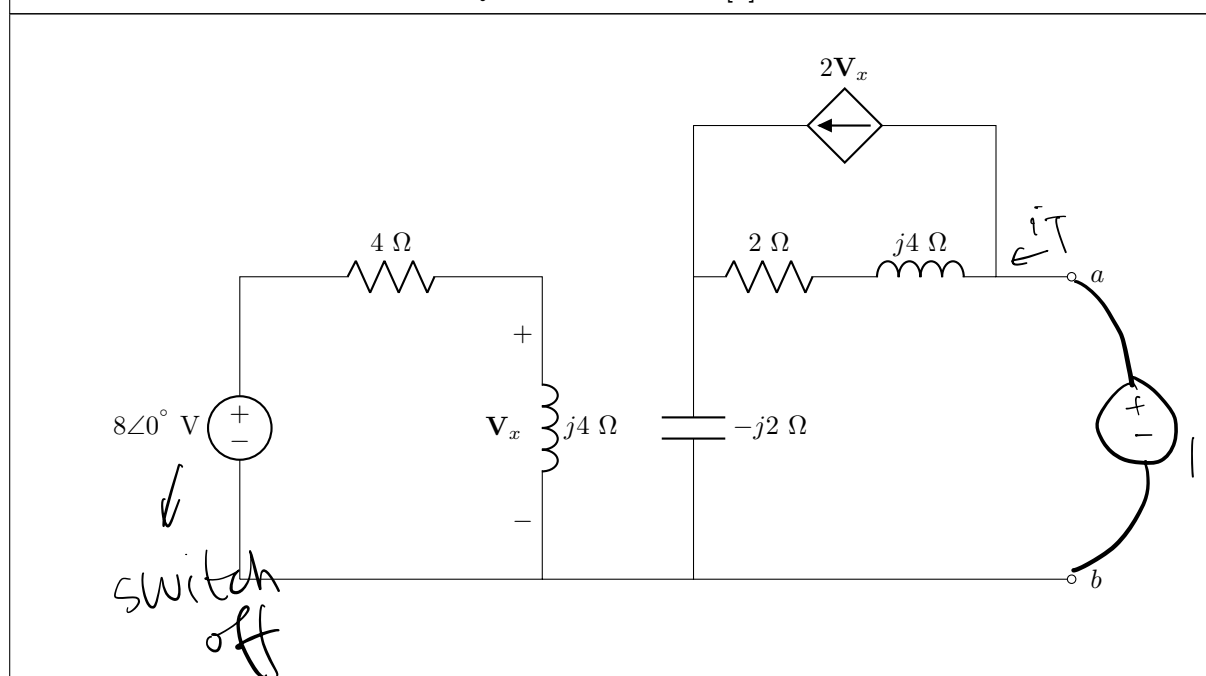


24 Find the contribution to I_x from the **independent** current source. [2] -3 A

$$-3 + I_x - 2I_x = 0$$

$$I_x = -3\text{ A}.$$

Questions 25 to 26 [4]



Find the Thévenin equivalent voltage V_{Th} and impedance Z_{Th} at terminals $a - b$.

25	V_{Th} [2]	$16 - j48 \text{ V}$
26	Z_{Th} [2]	$2 + j2 \Omega$

(25) $V_x = 8 \cdot \frac{j4}{4 + j4} = 4 + j4$

Circuit on the right; source transformation:

$$V_{Th} = -2V_x - (2 + j4) = -2(4 + j4)(2 + j4) \\ = 16 - j48 \text{ V}$$

(26) Add 1 V test voltage, switch off $8\angle 0^\circ$ to 0V.
 $V_x = 0, \quad i_T = \frac{1}{2 + j2} \Rightarrow Z_{Th} = \frac{1}{i_T} = 2 + j2 \Omega$

Total Section 3: [16]

Grand Total : [46]